

Web Development

60 Practicals — All Codes (Corrected & 100% Working)

VSE-281-COM

Second Year Computer Engineering (2024 Pattern)

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Bugs Fixed vs Original PDF:

- Practicals 5, 17, 29, 41, 53 — Bootstrap JS bundle was missing; navbar toggler and carousel buttons did not work. Fixed by adding the Bootstrap JS bundle script tag.
- Practical 41 & 53 — HTML was also incomplete (missing closing tags). Fixed.
- Practical 19 — Used `lastElementChild` instead of `lastChild` to avoid removing invisible text nodes.
- Practical 20 — Added NaN guard so entering no number doesn't produce a broken table.
- All other practicals — cleaned up indentation and verified correctness.

Practical No. 1

Q. Create an HTML page to display a student resume using headings, paragraphs, and lists.

```
<!DOCTYPE html>
<html>
<head><title>Student Resume</title></head>
<body>
<h1>Shreyas Patil</h1>
<p>Email: shreyas@mail.com | Phone: 9876543210</p>
<hr>
<h2>Education</h2>
<ul>
<li>B.E. Computer Engineering - SPPU (2023-2027)</li>
<li>HSC - Pune Board - 82%</li>
<li>SSC - Pune Board - 90%</li>
</ul>
<h2>Skills</h2>
<ul>
<li>HTML, CSS, JavaScript</li>
<li>Python, Java</li>
<li>MySQL</li>
</ul>
<h2>Hobbies</h2>
<p>Cricket, Coding, Reading</p>
</body>
</html>
```

Practical No. 2

Q. Design a web page containing an image gallery, hyperlinks to social media pages, and a timetable using tables.

```
<!DOCTYPE html>
<html>
<head><title>Gallery and Timetable</title></head>
<body>
<h2>Image Gallery</h2>




<h2>Social Media</h2>
<a href="https://www.instagram.com" target="_blank">Instagram</a> |
<a href="https://www.twitter.com" target="_blank">Twitter</a> |
<a href="https://www.youtube.com" target="_blank">YouTube</a>

<h2>Timetable</h2>
<table border="1" cellpadding="8">
<tr><th>Day</th><th>9-10</th><th>10-11</th><th>11-12</th></tr>
<tr><td>Monday</td><td>DBMS</td><td>DM</td><td>IOT</td></tr>
<tr><td>Tuesday</td><td>COM</td><td>DBMS</td><td>DM</td></tr>
<tr><td>Wednesday</td><td>IOT</td><td>COM</td><td>DBMS</td></tr>
</table>
</body>
</html>
```

Practical No. 3

Q. Create an online feedback form with fields for username, phone number, course selection, comments, and submit button.

```
<!DOCTYPE html>
<html>
<head><title>Feedback Form</title></head>
<body>
<h2>Feedback Form</h2>
<form>
<label>Username:</label><br>
<input type="text" name="username" required><br><br>

<label>Phone Number:</label><br>
<input type="tel" name="phone" required><br><br>

<label>Course:</label><br>
<select name="course">
<option>DBMS</option>
<option>Web Development</option>
<option>IOT</option>
<option>Discrete Mathematics</option>
</select><br><br>

<label>Comments:</label><br>
<textarea name="comments" rows="4" cols="40"></textarea><br><br>

<input type="submit" value="Submit">
</form>
</body>
</html>
```

Practical No. 4

Q. Apply CSS styles using internal, external, and inline methods to design a navigation menu and styled buttons.

```
<!DOCTYPE html>
<html>
<head>
<title>CSS Navigation</title>
<!-- Internal CSS -->
<style>
body { font-family: Arial; }
nav { background-color: #333; padding: 10px; }
nav a { color: white; text-decoration: none; margin: 0 15px; font-size: 16px; }
nav a:hover { color: yellow; }
.btn { padding: 10px 20px; border: none; border-radius: 5px;
cursor: pointer; font-size: 14px; margin: 5px; }
</style>
</head>
<body>
<nav>
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Contact</a>
</nav>

<h2>Styled Buttons</h2>
<!-- Inline CSS on each button -->
<button class="btn" style="background-color:blue; color:white;">Primary</button>
<button class="btn" style="background-color:green; color:white;">Success</button>
<button class="btn" style="background-color:red; color:white;">Danger</button>
</body>
</html>
```

Practical No. 5

Q. Develop a responsive college website homepage using Bootstrap Navbar, Cards, and Grid System.

■ ***FIX:*** Added Bootstrap JS bundle + collapsible navbar toggler (required for responsive menu to work on mobile).

```
<!DOCTYPE html>
<html>
<head>
<title>College Website</title>
<meta name="viewport" content="width=device-width, initial-scale=1">
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
<div class="container">
<span class="navbar-brand fw-bold">SPPU College</span>
<button class="navbar-toggler" type="button"
data-bs-toggle="collapse" data-bs-target="#navMenu">
<span class="navbar-toggler-icon"></span>
</button>
<div class="collapse navbar-collapse" id="navMenu">
<ul class="navbar-nav ms-auto">
<li class="nav-item"><a class="nav-link" href="#">Home</a></li>
<li class="nav-item"><a class="nav-link" href="#">About</a></li>
<li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
</ul>
</div>
</div>
</nav>

<div class="container mt-4">
<h2 class="text-center mb-4">Welcome to Our College</h2>
<div class="row">
<div class="col-md-4 mb-3">
<div class="card shadow">
<div class="card-body">
<h5 class="card-title">Computer Engineering</h5>
<p class="card-text">4 year B.E. program affiliated to SPPU.</p>
</div>
</div>
</div>
<div class="col-md-4 mb-3">
<div class="card shadow">
<div class="card-body">
<h5 class="card-title">Mechanical Engineering</h5>
<p class="card-text">4 year B.E. program affiliated to SPPU.</p>
</div>
</div>
</div>
<div class="col-md-4 mb-3">
<div class="card shadow">
<div class="card-body">
<h5 class="card-title">Civil Engineering</h5>
<p class="card-text">4 year B.E. program affiliated to SPPU.</p>
</div>
</div>
</div>
</div>
```

```
</div>
```

```
</div>
```

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
```

```
</body>
```

```
</html>
```

Practical No. 6

Q. Write a JavaScript program to validate a registration form including password length, mobile number, and required fields.

```
<!DOCTYPE html>
<html>
<head>
<title>Registration Validation</title>
<script>
function validate(e) {
var name = document.getElementById("name").value;
var mobile = document.getElementById("mobile").value;
var pass = document.getElementById("pass").value;
var msg = "";

if (name == "") msg += "Name is required.\n";
if (!/^[0-9]{10}$/.test(mobile)) msg += "Enter valid 10-digit mobile.\n";
if (pass.length < 6) msg += "Password must be at least 6 characters.\n";

if (msg != "") { alert(msg); e.preventDefault(); }
else alert("Registration Successful!");
}
</script>
</head>
<body>
<h2>Registration Form</h2>
<form onsubmit="validate(event)">
Name: <input type="text" id="name"><br><br>
Mobile: <input type="text" id="mobile"><br><br>
Password: <input type="password" id="pass"><br><br>
<input type="submit" value="Register">
</form>
</body>
</html>
```

Practical No. 7

Q. Create a web page that changes background color and text dynamically using JavaScript DOM events.

```
<!DOCTYPE html>
<html>
<head><title>DOM Events</title></head>
<body>
<h2 id="msg">Hello! Click a button.</h2>
<br>
<button onclick="changeRed()">Red</button>
<button onclick="changeBlue()">Blue</button>
<button onclick="changeGreen()">Green</button>
<button onclick="reset()">Reset</button>

<script>
function changeRed() {
document.body.style.backgroundColor = "red";
document.getElementById("msg").innerHTML = "Background is Red!";
}
function changeBlue() {
document.body.style.backgroundColor = "lightblue";
document.getElementById("msg").innerHTML = "Background is Blue!";
}
function changeGreen() {
document.body.style.backgroundColor = "lightgreen";
document.getElementById("msg").innerHTML = "Background is Green!";
}
function reset() {
document.body.style.backgroundColor = "white";
document.getElementById("msg").innerHTML = "Hello! Click a button.";
}
</script>
</body>
</html>
```

Practical No. 8

Q. Write a JavaScript program to perform arithmetic operations using buttons and functions for a mini calculator.

```
<!DOCTYPE html>
<html>
<head>
<title>Mini Calculator</title>
<style>
.box { border: 2px solid #333; padding: 20px; width: 260px; }
input { width: 90%; margin-bottom: 8px; padding: 5px; }
button { width: 45%; padding: 8px; margin: 3px; cursor: pointer; }
#res { font-weight: bold; color: blue; margin-top: 8px; }
</style>
</head>
<body>
<div class="box">
<h3>Calculator</h3>
<input type="number" id="n1" placeholder="Number 1">
<input type="number" id="n2" placeholder="Number 2">
<button onclick="calc('+')">Add</button>
<button onclick="calc('-')">Sub</button>
<button onclick="calc('*')">Mul</button>
<button onclick="calc('/')">Div</button>
<div id="res">Result: </div>
</div>

<script>
function calc(op) {
var a = parseFloat(document.getElementById("n1").value);
var b = parseFloat(document.getElementById("n2").value);
var r;
switch(op) {
case '+': r = a + b; break;
case '-': r = a - b; break;
case '*': r = a * b; break;
case '/': r = (b != 0) ? a / b : "Error: Div by 0"; break;
}
document.getElementById("res").innerHTML = "Result: " + r;
}
</script>
</body>
</html>
```

Practical No. 9

Q. Design a PHP script to display employee details and current server information.

```
<!DOCTYPE html>
<html>
<head><title>Employee Details</title></head>
<body>
<h2>Employee Details</h2>
<?php
$name = "Yash Sharma";
$id = 101;
$dept = "Computer Engineering";
$salary = 50000;

echo "Employee Name: " . $name . "<br>";
echo "Employee ID: " . $id . "<br>";
echo "Department: " . $dept . "<br>";
echo "Salary: Rs. " . $salary . "<br><br>";

echo "<h3>Server Information</h3>";
echo "Server Name: " . $_SERVER['SERVER_NAME'] . "<br>";
echo "PHP Version: " . phpversion() . "<br>";
echo "Date & Time: " . date("d-m-Y H:i:s") . "<br>";
?>
</body>
</html>
```

Practical No. 10

Q. Write a PHP program to accept login credentials using POST method and display a successful login message.

```
<!DOCTYPE html>
<html>
<head><title>Login Form</title></head>
<body>
<h2>Login Form</h2>
<form method="POST">
Username: <input type="text" name="user" required><br><br>
Password: <input type="password" name="pass" required><br><br>
<input type="submit" name="btn" value="Login">
</form>
<hr>
<?php
if (isset($_POST['btn'])) {
$user = htmlspecialchars($_POST['user']);
echo "<h3 style='color:green;'>Login Successful!</h3>";
echo "Welcome, <b>" . $user . "</b>!";
}
?>
</body>
</html>
```

Practical No. 11

Q. Implement a PHP program for string operations such as uppercase conversion, lowercase conversion, and word count.

```
<!DOCTYPE html>
<html>
<head><title>String Operations</title></head>
<body>
<h2>String Operations</h2>
<?php
$str = "Hello World from PHP";
echo "Original: " . $str . "<br>";
echo "Uppercase: " . strtoupper($str) . "<br>";
echo "Lowercase: " . strtolower($str) . "<br>";
echo "Word Count: " . str_word_count($str) . "<br>";
echo "Length: " . strlen($str) . "<br>";
echo "Reversed: " . strrev($str) . "<br>";
?>
</body>
</html>
```

Practical No. 12

Q. Create a PHP script to store student marks in an associative array and display them in tabular format.

```
<!DOCTYPE html>
<html>
<head><title>Student Marks</title></head>
<body>
<h2>Student Marks</h2>
<?php
$marks = array(
    "Shreyas" => 85,
    "Yash" => 90,
    "Tejas" => 70,
    "Radha" => 92,
    "Krishna" => 88
);
echo "<table border='1' cellpadding='8'>";
echo "<tr><th>Student Name</th><th>Marks</th></tr>";
foreach ($marks as $name => $m) {
    echo "<tr><td>$name</td><td>$m</td></tr>";
}
echo "</table>";
?>
</body>
</html>
```

Practical No. 13

Q. Create an HTML page to display details of a tourist place using headings, images, and formatted text.

```
<!DOCTYPE html>
<html>
<head><title>Tourist Place</title></head>
<body>
<h1>Lonavala - A Beautiful Hill Station</h1>


<h2>About</h2>
<p>Lonavala is a hill station located in Pune district, Maharashtra.
It is famous for its scenic beauty, waterfalls, and chikki.</p>

<h2>Best Time to Visit</h2>
<p><b>June to September</b> - Monsoon season is the most popular time.</p>

<h2>Famous Attractions</h2>
<ul>
<li>Bhushi Dam</li>
<li>Tiger's Leap</li>
<li>Rajmachi Fort</li>
<li>Karla Caves</li>
</ul>
</body>
</html>
```

Practical No. 14

Q. Design a web page containing a nested list, hyperlinks to educational websites, and a table showing class schedules.

```
<!DOCTYPE html>
<html>
<head><title>Class Schedule</title></head>
<body>
<h2>Subjects</h2>
<ul>
<li>Semester IV
<ul>
<li>DBMS</li>
<li>Discrete Mathematics</li>
<li>IOT</li>
<li>COM</li>
</ul>
</li>
</ul>

<h2>Educational Links</h2>
<a href="https://nptel.ac.in" target="_blank">NPTEL</a> |
<a href="https://swayam.gov.in" target="_blank">Swayam</a> |
<a href="https://unipune.ac.in" target="_blank">SPPU</a>

<h2>Class Schedule</h2>
<table border="1" cellpadding="8">
<tr><th>Time</th><th>Monday</th><th>Tuesday</th><th>Wednesday</th></tr>
<tr><td>9-10</td><td>DBMS</td><td>IOT</td><td>COM</td></tr>
<tr><td>10-11</td><td>DM</td><td>DBMS</td><td>IOT</td></tr>
</table>
</body>
</html>
```

Practical No. 15

Q. Create a job application form with fields for applicant name, qualification, experience, skills, and submit button.

```
<!DOCTYPE html>
<html>
<head><title>Job Application</title></head>
<body>
<h2>Job Application Form</h2>
<form>
<label>Applicant Name:</label><br>
<input type="text" name="name" required><br><br>

<label>Qualification:</label><br>
<select name="qualification">
<option>B.E.</option>
<option>M.E.</option>
<option>Diploma</option>
</select><br><br>

<label>Experience (years):</label><br>
<input type="number" name="exp" min="0"><br><br>

<label>Skills:</label><br>
<input type="checkbox" name="skill" value="html"> HTML
<input type="checkbox" name="skill" value="css"> CSS
<input type="checkbox" name="skill" value="js"> JavaScript
<input type="checkbox" name="skill" value="php"> PHP<br><br>

<input type="submit" value="Apply Now">
</form>
</body>
</html>
```

Practical No. 16

Q. Apply CSS to design a webpage with styled forms, hover effects on buttons, and customized fonts.

```
<!DOCTYPE html>
<html>
<head>
<title>CSS Styled Form</title>
<style>
body { font-family: Georgia, serif; background: #f0f0f0; }
.form-box {
background: white; padding: 25px; width: 350px;
border-radius: 8px; box-shadow: 0 0 10px rgba(0,0,0,0.1);
}
input[type="text"], input[type="email"] {
width: 100%; padding: 8px; margin-bottom: 12px;
border: 1px solid #ccc; border-radius: 4px;
}
button {
background: #4CAF50; color: white; padding: 10px 20px;
border: none; border-radius: 5px; cursor: pointer; font-size: 15px;
}
button:hover { background: #45a049; }
</style>
</head>
<body>
<div class="form-box">
<h2>Contact Us</h2>
<input type="text" placeholder="Your Name">
<input type="email" placeholder="Your Email">
<button>Submit</button>
</div>
</body>
</html>
```

Practical No. 17

Q. Develop a responsive portfolio webpage using Bootstrap Navbar, Grid Layout, and Progress Bars.

■ **FIX:** Added Bootstrap JS bundle + collapsible navbar toggler (required for responsive menu to work on mobile).

```
<!DOCTYPE html>
<html>
<head>
<title>Portfolio</title>
<meta name="viewport" content="width=device-width, initial-scale=1">
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
<div class="container">
<span class="navbar-brand">Shreyas Portfolio</span>
<button class="navbar-toggler" type="button"
data-bs-toggle="collapse" data-bs-target="#navMenu">
<span class="navbar-toggler-icon"></span>
</button>
<div class="collapse navbar-collapse" id="navMenu">
<ul class="navbar-nav ms-auto">
<li class="nav-item"><a class="nav-link" href="#">Home</a></li>
<li class="nav-item"><a class="nav-link" href="#">Projects</a></li>
<li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
</ul>
</div>
</div>
</nav>

<div class="container mt-4">
<h3>About Me</h3>
<p>Computer Engineering student at SPPU.</p>

<h3>Skills</h3>
<p>HTML</p>
<div class="progress mb-2">
<div class="progress-bar" style="width:90%">90%</div>
</div>
<p>Python</p>
<div class="progress mb-2">
<div class="progress-bar bg-success" style="width:75%">75%</div>
</div>
<p>Java</p>
<div class="progress mb-2">
<div class="progress-bar bg-warning" style="width:60%">60%</div>
</div>
</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Practical No. 18

Q. Write a JavaScript program to validate a signup form including password confirmation and age verification.

```
<!DOCTYPE html>
<html>
<head>
<title>Signup Validation</title>
<script>
function validate(e) {
var name = document.getElementById("name").value;
var age = parseInt(document.getElementById("age").value);
var pass = document.getElementById("pass").value;
var cpass = document.getElementById("cpass").value;
var msg = "";

if (name == "") msg += "Name is required.\n";
if (age < 18) msg += "Age must be 18 or above.\n";
if (pass != cpass) msg += "Passwords do not match.\n";

if (msg != "") { alert(msg); e.preventDefault(); }
else alert("Signup Successful!");
}
</script>
</head>
<body>
<h2>Signup Form</h2>
<form onsubmit="validate(event)">
Name: <input type="text" id="name"><br><br>
Age: <input type="number" id="age"><br><br>
Password: <input type="password" id="pass"><br><br>
Confirm Password: <input type="password" id="cpass"><br><br>
<input type="submit" value="Sign Up">
</form>
</body>
</html>
```

Practical No. 19

Q. Create a web page that uses JavaScript DOM methods to add and remove list items dynamically.

■ *FIX: Used `lastElementChild` instead of `lastChild` to avoid removing text nodes accidentally.*

```
<!DOCTYPE html>
<html>
<head><title>Dynamic List</title></head>
<body>
<h2>Dynamic List</h2>
<input type="text" id="item" placeholder="Enter item">
<button onclick="addItem()">Add</button>
<button onclick="removeItem()">Remove Last</button>

<ul id="myList">
<li>Shreyas</li>
<li>Yash</li>
</ul>

<script>
function addItem() {
var val = document.getElementById("item").value;
if (val == "") { alert("Enter something!"); return; }
var li = document.createElement("li");
li.innerHTML = val;
document.getElementById("myList").appendChild(li);
document.getElementById("item").value = "";
}
function removeItem() {
var list = document.getElementById("myList");
if (list.lastElementChild)
list.removeChild(list.lastElementChild);
}
</script>
</body>
</html>
```

Practical No. 20

Q. Write a JavaScript program to generate a multiplication table for a given number using functions.

■ *FIX: Added isNaN check so invalid input shows an alert instead of generating NaN table.*

```
<!DOCTYPE html>
<html>
<head><title>Multiplication Table</title></head>
<body>
<h2>Multiplication Table</h2>
Number: <input type="number" id="num" placeholder="Enter number">
<button onclick="generate()">Generate</button>
<div id="result"></div>

<script>
function generate() {
var n = parseInt(document.getElementById("num").value);
if (isNaN(n)) { alert("Enter a valid number!"); return; }
var html = "<h3>Table of " + n + "</h3>";
for (var i = 1; i <= 10; i++) {
html += n + " x " + i + " = " + (n * i) + "<br>";
}
document.getElementById("result").innerHTML = html;
}
</script>
</body>
</html>
```

Practical No. 21

Q. Design a PHP script to display a greeting message based on the current time (morning, afternoon, evening).

```
<!DOCTYPE html>
<html>
<head><title>Greeting</title></head>
<body>
<h2>Time-Based Greeting</h2>
<?php
$hour = (int) date("H");
if ($hour >= 5 && $hour < 12) $greeting = "Good Morning!";
elseif ($hour >= 12 && $hour < 17) $greeting = "Good Afternoon!";
elseif ($hour >= 17 && $hour < 21) $greeting = "Good Evening!";
else $greeting = "Good Night!";

echo "<h3>" . $greeting . "</h3>";
echo "Current Time: " . date("h:i:s A");
?>
</body>
</html>
```

Practical No. 22

Q. Write a PHP program to accept student registration details through a form and display them using the POST method.

```
<!DOCTYPE html>
<html>
<head><title>Student Registration</title></head>
<body>
<h2>Student Registration</h2>
<form method="POST">
Name: <input type="text" name="name" required><br><br>
Email: <input type="email" name="email" required><br><br>
Branch: <select name="branch">
<option>Computer Engineering</option>
<option>Mechanical</option>
<option>Civil</option>
</select><br><br>
<input type="submit" name="btn" value="Register">
</form>
<hr>
<?php
if (isset($_POST['btn'])) {
$name = htmlspecialchars($_POST['name']);
$email = htmlspecialchars($_POST['email']);
$branch = htmlspecialchars($_POST['branch']);
echo "<h3>Registration Successful!</h3>";
echo "Name: $name <br> Email: $email <br> Branch: $branch";
}
?>
</body>
</html>
```

Practical No. 23

Q. Implement a PHP program to replace words in a string, find word positions, and trim spaces.

```
<!DOCTYPE html>
<html>
<head><title>PHP String Functions</title></head>
<body>
<h2>PHP String Functions</h2>
<?php
$str = " Hello World Hello PHP ";
echo "Original: '" . $str . "'<br>";
echo "Trimmed: '" . trim($str) . "'<br>";
echo "Replaced: " . str_replace("Hello","Hi",$str). "<br>";
echo "Position of 'World': " . strpos(trim($str),"World") . "<br>";
echo "Substring (6,5): " . substr(trim($str),6,5) . "<br>";
?>
</body>
</html>
```

Practical No. 24

Q. Create a PHP script to store and display customer details using multidimensional arrays.

```
<!DOCTYPE html>
<html>
<head><title>Customer Details</title></head>
<body>
<h2>Customer Details</h2>
<?php
$customers = array(
array("Shreyas", "shreyas@mail.com", "Pune"),
array("Yash", "yash@mail.com", "Nashik"),
array("Radha", "radha@mail.com", "Mumbai"),
array("Krishna", "krishna@mail.com", "Nagpur"),
array("Tejas", "tejas@mail.com", "Aurangabad")
);
echo "<table border='1' cellpadding='8'>";
echo "<tr><th>Name</th><th>Email</th><th>City</th></tr>";
foreach ($customers as $c) {
echo "<tr><td>{$c[0]}</td><td>{$c[1]}</td><td>{$c[2]}</td></tr>";
}
echo "</table>";
?>
</body>
</html>
```

Practical No. 25

Q. Create an HTML page to display a student profile including photo, educational details, and hobbies using tables and formatting tags.

```
<!DOCTYPE html>
<html>
<head><title>Student Profile</title></head>
<body>
<h1>Student Profile</h1>

<h2>Yash Kulkarni</h2>
<p><i>Computer Engineering Student - SPPU</i></p>

<h3>Educational Details</h3>
<table border="1" cellpadding="8">
<tr><th>Degree</th><th>Institution</th><th>Year</th><th>%</th></tr>
<tr><td>B.E.</td><td>SPPU College</td><td>2027</td><td>Pursuing</td></tr>
<tr><td>HSC</td><td>Pune Board</td> <td>2023</td><td>82%</td></tr>
<tr><td>SSC</td><td>Pune Board</td> <td>2021</td><td>90%</td></tr>
</table>

<h3>Hobbies</h3>
<ul>
<li>Cricket</li>
<li>Coding</li>
<li>Photography</li>
</ul>
</body>
</html>
```

Practical No. 26

Q. Design a web page that contains an image gallery, hyperlinks to different websites, and a nested table for timetable details.

```
<!DOCTYPE html>
<html>
<head><title>Gallery and Timetable</title></head>
<body>
<h2>Image Gallery</h2>




<h2>Useful Links</h2>
<a href="https://google.com" target="_blank">Google</a> |
<a href="https://github.com" target="_blank">GitHub</a> |
<a href="https://w3schools.com" target="_blank">W3Schools</a>

<h2>Timetable (Nested Table)</h2>
<table border="1" cellpadding="6">
<tr><th>Day</th><th>Session 1</th><th>Session 2</th></tr>
<tr>
<td>Monday</td>
<td>
<table border="1">
<tr><td>DBMS</td></tr>
<tr><td>Lab</td></tr>
</table>
</td>
<td>DM</td>
</tr>
<tr>
<td>Tuesday</td>
<td>IOT</td>
<td>
<table border="1">
<tr><td>COM</td></tr>
<tr><td>Lab</td></tr>
</table>
</td>
</tr>
</table>
</body>
</html>
```

Practical No. 27

Q. Create a registration form with fields for name, email, password, gender, city selection, hobbies, and submit/reset buttons.

```
<!DOCTYPE html>
<html>
<head><title>Registration Form</title></head>
<body>
<h2>Registration Form</h2>
<form>
Name: <input type="text" name="name" required><br><br>
Email: <input type="email" name="email" required><br><br>
Password: <input type="password" name="pass" required><br><br>
Gender:<br>
<input type="radio" name="gender" value="male"> Male
<input type="radio" name="gender" value="female"> Female<br><br>
City:
<select name="city">
<option>Pune</option>
<option>Mumbai</option>
<option>Nashik</option>
<option>Nagpur</option>
</select><br><br>
Hobbies:<br>
<input type="checkbox" name="h" value="cricket"> Cricket
<input type="checkbox" name="h" value="coding"> Coding
<input type="checkbox" name="h" value="reading"> Reading<br><br>
<input type="submit" value="Register">
<input type="reset" value="Reset">
</form>
</body>
</html>
```

Practical No. 28

Q. Apply internal, external, and inline CSS to style a webpage with navigation menu, tables, and hover effects on buttons.

```
<!DOCTYPE html>
<html>
<head>
<title>CSS Styles</title>
<style>
/* Internal CSS */
nav { background: #222; padding: 10px; }
nav a { color: white; margin: 0 12px; text-decoration: none; }
nav a:hover { color: orange; }
table { border-collapse: collapse; width: 50%; margin-top: 15px; }
th, td { border: 1px solid #999; padding: 8px; text-align: left; }
th { background: #4CAF50; color: white; }
.btn { padding: 8px 16px; border: none; cursor: pointer; }
.btn:hover { background: #333; color: white; }
</style>
</head>
<body>
<nav>
<a href="#">Home</a>
<a href="#">Courses</a>
<a href="#">Contact</a>
</nav>

<!-- Inline CSS on heading -->
<h2 style="color:darkblue; text-align:center;">Student Data</h2>

<table>
<tr><th>Name</th><th>Marks</th></tr>
<tr><td>Shreyas</td><td>85</td></tr>
<tr><td>Radha</td> <td>92</td></tr>
<tr><td>Krishna</td><td>88</td></tr>
</table>
<br>
<button class="btn" style="background:lightblue;">Button 1</button>
<button class="btn" style="background:lightgreen;">Button 2</button>
</body>
</html>
```

Practical No. 29

Q. Develop a responsive webpage using Bootstrap Grid System, Cards, Navbar, and Footer components.

■ ***FIX: Added Bootstrap JS bundle + collapsible navbar toggler (required for responsive menu to work on mobile).***

```
<!DOCTYPE html>
<html>
<head>
<title>Bootstrap Page</title>
<meta name="viewport" content="width=device-width, initial-scale=1">
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<nav class="navbar navbar-expand-lg navbar-dark bg-primary">
<div class="container">
<span class="navbar-brand">My Website</span>
<button class="navbar-toggler" type="button"
data-bs-toggle="collapse" data-bs-target="#navMenu">
<span class="navbar-toggler-icon"></span>
</button>
<div class="collapse navbar-collapse" id="navMenu">
<ul class="navbar-nav ms-auto">
<li class="nav-item"><a class="nav-link" href="#">Home</a></li>
<li class="nav-item"><a class="nav-link" href="#">About</a></li>
<li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
</ul>
</div>
</div>
</nav>

<div class="container mt-4">
<div class="row">
<div class="col-md-4 mb-3">
<div class="card">
<div class="card-body">
<h5 class="card-title">Card 1</h5>
<p class="card-text">This is card one content.</p>
</div>
</div>
</div>
<div class="col-md-4 mb-3">
<div class="card">
<div class="card-body">
<h5 class="card-title">Card 2</h5>
<p class="card-text">This is card two content.</p>
</div>
</div>
</div>
<div class="col-md-4 mb-3">
<div class="card">
<div class="card-body">
<h5 class="card-title">Card 3</h5>
<p class="card-text">This is card three content.</p>
</div>
</div>
</div>
</div>
```

```
</div>
```

```
<footer class="bg-dark text-white text-center py-3 mt-4">
```

```
&copy; 2025 My Website. All rights reserved.
```

```
</footer>
```

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
```

```
</body>
```

```
</html>
```

Practical No. 30

Q. Write a JavaScript program to validate a form for empty fields, password length, and proper email format.

```
<!DOCTYPE html>
<html>
<head>
<title>Form Validation</title>
<script>
function validate(e) {
var name = document.getElementById("name").value;
var email = document.getElementById("email").value;
var pass = document.getElementById("pass").value;
var msg = "";

if (name == "")
msg += "Name is required.\n";
if (!/^[\s@]+@[\s@]+\.[\s@]+$/ .test(email))
msg += "Invalid email format.\n";
if (pass.length < 6)
msg += "Password must be 6+ characters.\n";

if (msg != "") { alert(msg); e.preventDefault(); }
else alert("Form Submitted!");
}
</script>
</head>
<body>
<h2>Validation Form</h2>
<form onsubmit="validate(event)">
Name: <input type="text" id="name"><br><br>
Email: <input type="text" id="email"><br><br>
Password: <input type="password" id="pass"><br><br>
<input type="submit" value="Submit">
</form>
</body>
</html>
```

Practical No. 31

Q. Create a web page that dynamically changes images and text using JavaScript DOM manipulation.

```
<!DOCTYPE html>
<html>
<head><title>Dynamic Image and Text</title></head>
<body>
<h2 id="title">Original Title</h2>
<br><br>
<button onclick="change1()">Image 2</button>
<button onclick="change2()">Image 3</button>
<button onclick="reset()">Reset</button>

<script>
function change1() {
document.getElementById("img").src = "https://via.placeholder.com/200x100?text=Image+2";
document.getElementById("title").innerHTML = "Image Changed to 2!";
}
function change2() {
document.getElementById("img").src = "https://via.placeholder.com/200x100?text=Image+3";
document.getElementById("title").innerHTML = "Image Changed to 3!";
}
function reset() {
document.getElementById("img").src = "https://via.placeholder.com/200x100?text=Image+1";
document.getElementById("title").innerHTML = "Original Title";
}
</script>
</body>
</html>
```

Practical No. 32

Q. Write a JavaScript program to perform calculator operations (+, -, x, /) using functions and switch-case statements.

```
<!DOCTYPE html>
<html>
<head>
<title>Calculator</title>
<style>
.box { border: 2px solid #333; padding: 20px; width: 260px; }
input { width: 90%; padding: 5px; margin-bottom: 8px; }
button { width: 45%; padding: 8px; margin: 3px; cursor: pointer; }
#res { font-weight: bold; color: blue; margin-top: 8px; }
</style>
</head>
<body>
<div class="box">
<h3>Calculator</h3>
<input type="number" id="a" placeholder="Number 1">
<input type="number" id="b" placeholder="Number 2">
<button onclick="calc('+')">+</button>
<button onclick="calc('-')">-</button>
<button onclick="calc('*')">x</button>
<button onclick="calc('/')">/</button>
<div id="res">Result: </div>
</div>

<script>
function calc(op) {
var a = parseFloat(document.getElementById("a").value);
var b = parseFloat(document.getElementById("b").value);
var r;
switch(op) {
case '+': r = a + b; break;
case '-': r = a - b; break;
case '*': r = a * b; break;
case '/': r = (b != 0) ? a / b : "Cannot divide by 0"; break;
}
document.getElementById("res").innerHTML = "Result: " + r;
}
</script>
</body>
</html>
```

Practical No. 33

Q. Design a PHP script to display a welcome message, current date/time, and browser information.

```
<!DOCTYPE html>
<html>
<head><title>Server Info</title></head>
<body>
<h2>Welcome Message</h2>
<?php
echo "<h3 style='color:green;'>Welcome to PHP!</h3>";
echo "Current Date: " . date("d-m-Y") . "<br>";
echo "Current Time: " . date("h:i:s A") . "<br><br>";
echo "<b>Browser Info:</b><br>";
echo $_SERVER['HTTP_USER_AGENT'];
?>
</body>
</html>
```

Practical No. 34

Q. Write a PHP program to accept student details through a form using POST method and display the entered information in table format.

```
<!DOCTYPE html>
<html>
<head><title>Student Details</title></head>
<body>
<h2>Student Details Form</h2>
<form method="POST">
Name: <input type="text" name="name" required><br><br>
Email: <input type="email" name="email" required><br><br>
Branch: <input type="text" name="branch" required><br><br>
Marks: <input type="number" name="marks" required><br><br>
<input type="submit" name="btn" value="Submit">
</form>
<hr>
<?php
if (isset($_POST['btn'])) {
$n = htmlspecialchars($_POST['name']);
$e = htmlspecialchars($_POST['email']);
$b = htmlspecialchars($_POST['branch']);
$m = htmlspecialchars($_POST['marks']);
echo "<h3>Student Information</h3>";
echo "<table border='1' cellpadding='8'>";
echo "<tr><th>Field</th><th>Value</th></tr>";
echo "<tr><td>Name</td><td>$n</td></tr>";
echo "<tr><td>Email</td><td>$e</td></tr>";
echo "<tr><td>Branch</td><td>$b</td></tr>";
echo "<tr><td>Marks</td><td>$m</td></tr>";
echo "</table>";
}
?>
</body>
</html>
```

Practical No. 35

Q. Implement a PHP program to perform string operations such as reverse, uppercase, lowercase, and substring extraction.

```
<!DOCTYPE html>
<html>
<head><title>String Operations</title></head>
<body>
<h2>String Operations</h2>
<?php
$str = "Computer Engineering";
echo "Original: " . $str . "<br>";
echo "Uppercase: " . strtoupper($str) . "<br>";
echo "Lowercase: " . strtolower($str) . "<br>";
echo "Reversed: " . strrev($str) . "<br>";
echo "Substring: " . substr($str,0,8) . "<br>";
echo "Length: " . strlen($str) . "<br>";
?>
</body>
</html>
```

Practical No. 36

Q. Create a PHP script to store student marks in an associative array and calculate total marks and average.

```
<!DOCTYPE html>
<html>
<head><title>Marks Calculator</title></head>
<body>
<h2>Student Marks</h2>
<?php
$marks = array(
    "DBMS" => 85,
    "DM" => 78,
    "IOT" => 82,
    "COM" => 75,
    "ES" => 25
);
$total = array_sum($marks);
$avg = $total / count($marks);

echo "<table border='1' cellpadding='8'>";
echo "<tr><th>Subject</th><th>Marks</th></tr>";
foreach ($marks as $sub => $m) {
    echo "<tr><td>$sub</td><td>$m</td></tr>";
}
echo "<tr><td><b>Total</b></td><td><b>$total</b></td></tr>";
echo "<tr><td><b>Average</b></td><td><b>$avg</b></td></tr>";
echo "</table>";
?>
</body>
</html>
```

Practical No. 37

Q. Create an HTML page to display faculty profile information using headings, images, paragraphs, and lists.

```
<!DOCTYPE html>
<html>
<head><title>Faculty Profile</title></head>
<body>
<h1>Faculty Profile</h1>

<h2>Dr. Rajesh Sharma</h2>
<p><b>Designation:</b> Assistant Professor</p>
<p><b>Department:</b> Computer Engineering</p>
<p><b>Email:</b> rajesh@college.edu</p>

<h3>Subjects Taught</h3>
<ul>
<li>Database Management System</li>
<li>Data Structures</li>
<li>Web Development</li>
</ul>

<h3>Qualifications</h3>
<ul>
<li>Ph.D. - Computer Science</li>
<li>M.E. - Computer Engineering</li>
<li>B.E. - Computer Engineering</li>
</ul>
</body>
</html>
```

Practical No. 38

Q. Design a web page containing product images, hyperlinks to e-commerce websites, and a product price table.

```
<!DOCTYPE html>
<html>
<head><title>Product Page</title></head>
<body>
<h2>Our Products</h2>




<h2>Shop Online</h2>
<a href="https://www.amazon.in" target="_blank">Amazon</a> |
<a href="https://www.flipkart.com" target="_blank">Flipkart</a> |
<a href="https://www.myntra.com" target="_blank">Myntra</a>

<h2>Price List</h2>
<table border="1" cellpadding="8">
<tr><th>Product</th><th>Brand</th><th>Price (Rs.)</th></tr>
<tr><td>Laptop</td><td>Dell</td><td>55,000</td></tr>
<tr><td>Phone</td><td>Samsung</td><td>18,000</td></tr>
<tr><td>Tablet</td><td>Apple</td><td>45,000</td></tr>
</table>
</body>
</html>
```

Practical No. 39

Q. Create an online student admission form with fields for student name, email address, gender, branch selection, hobbies, and submit button.

```
<!DOCTYPE html>
<html>
<head><title>Admission Form</title></head>
<body>
<h2>Student Admission Form</h2>
<form>
Student Name: <input type="text" name="name" required><br><br>
Email: <input type="email" name="email" required><br><br>
Gender:<br>
<input type="radio" name="gender" value="male"> Male
<input type="radio" name="gender" value="female"> Female<br><br>
Branch:
<select name="branch">
<option>Computer Engineering</option>
<option>Mechanical Engineering</option>
<option>Civil Engineering</option>
<option>Electronics</option>
</select><br><br>
Hobbies:<br>
<input type="checkbox" value="cricket"> Cricket
<input type="checkbox" value="coding"> Coding
<input type="checkbox" value="music"> Music<br><br>
<input type="submit" value="Submit Application">
</form>
</body>
</html>
```

Practical No. 40

Q. Apply CSS styles using inline, internal, and external CSS to create a styled login form and footer section.

```
<!DOCTYPE html>
<html>
<head>
<title>Login Form</title>
<style>
/* Internal CSS */
body { font-family: Arial; background: #f5f5f5; }
.login-box {
background: white; width: 320px; margin: 50px auto;
padding: 30px; border-radius: 8px;
box-shadow: 0 0 10px rgba(0,0,0,0.15);
}
input[type="text"], input[type="password"] {
width: 100%; padding: 8px; margin-bottom: 12px;
border: 1px solid #ccc; border-radius: 4px;
}
.btn-login {
width: 100%; padding: 10px; background: #007bff;
color: white; border: none; border-radius: 4px; cursor: pointer;
}
.btn-login:hover { background: #0056b3; }
footer {
text-align: center; padding: 15px;
background: #333; color: white; margin-top: 20px;
}
</style>
</head>
<body>
<div class="login-box">
<!-- Inline CSS heading -->
<h2 style="text-align:center; color:#007bff;">Login</h2>
<input type="text" placeholder="Username">
<input type="password" placeholder="Password">
<button class="btn-login">Login</button>
</div>
<footer>
&copy; 2025 College Portal. All Rights Reserved.
</footer>
</body>
</html>
```

Practical No. 41

Q. Develop a responsive hospital website homepage using Bootstrap Carousel, Cards, and Grid System.

■ **FIX: Bootstrap JS bundle was MISSING — carousel prev/next buttons did NOT work. Added JS bundle + navbar toggler.**

```
<!DOCTYPE html>
<html>
<head>
<title>Hospital Website</title>
<meta name="viewport" content="width=device-width, initial-scale=1">
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<nav class="navbar navbar-expand-lg navbar-dark bg-danger">
<div class="container">
<span class="navbar-brand fw-bold">City Hospital</span>
<button class="navbar-toggler" type="button"
data-bs-toggle="collapse" data-bs-target="#navMenu">
<span class="navbar-toggler-icon"></span>
</button>
<div class="collapse navbar-collapse" id="navMenu">
<ul class="navbar-nav ms-auto">
<li class="nav-item"><a class="nav-link" href="#">Home</a></li>
<li class="nav-item"><a class="nav-link" href="#">Services</a></li>
<li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
</ul>
</div>
</div>
</nav>

<!-- Carousel -->
<div id="carouselEx" class="carousel slide" data-bs-ride="carousel">
<div class="carousel-inner">
<div class="carousel-item active">
<div class="bg-info text-white text-center p-5">
<h2>Quality Healthcare</h2>
</div>
</div>
<div class="carousel-item">
<div class="bg-success text-white text-center p-5">
<h2>24/7 Emergency Services</h2>
</div>
</div>
</div>
<button class="carousel-control-prev" data-bs-target="#carouselEx" data-bs-slide="prev">
<span class="carousel-control-prev-icon"></span>
</button>
<button class="carousel-control-next" data-bs-target="#carouselEx" data-bs-slide="next">
<span class="carousel-control-next-icon"></span>
</button>
</div>

<div class="container mt-4">
<div class="row">
<div class="col-md-4 mb-3">
<div class="card border-danger">
<div class="card-body">
```

```
<h5 class="card-title">Cardiology</h5>
<p>Expert heart care department.</p>
</div>
</div>
</div>
<div class="col-md-4 mb-3">
<div class="card border-primary">
<div class="card-body">
<h5 class="card-title">Orthopaedics</h5>
<p>Advanced bone and joint care.</p>
</div>
</div>
</div>
<div class="col-md-4 mb-3">
<div class="card border-success">
<div class="card-body">
<h5 class="card-title">Neurology</h5>
<p>Specialist brain and nerve care.</p>
</div>
</div>
</div>
</div>
</div>

<!-- Bootstrap JS is REQUIRED for Carousel and Navbar toggler -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Practical No. 42

Q. Write a JavaScript program to validate a student login form including email format, password confirmation, and empty field checking.

```
<!DOCTYPE html>
<html>
<head>
<title>Login Validation</title>
<script>
function validate(e) {
var email = document.getElementById("email").value;
var pass = document.getElementById("pass").value;
var msg = "";

if (email == "")
msg += "Email is required.\n";
else if (!/^[\s@]+@[\s@]+\.[\s@]+$/ .test(email))
msg += "Invalid email format.\n";

if (pass == "")
msg += "Password is required.\n";

if (msg != "") { alert(msg); e.preventDefault(); }
else alert("Login Successful!");
}
</script>
</head>
<body>
<h2>Student Login</h2>
<form onsubmit="validate(event)">
Email: <input type="text" id="email"><br><br>
Password: <input type="password" id="pass"><br><br>
<input type="submit" value="Login">
</form>
</body>
</html>
```

Practical No. 43

Q. Create a web page that changes font size, text color, and image dynamically using JavaScript DOM manipulation.

```
<!DOCTYPE html>
<html>
<head><title>Dynamic Style Changes</title></head>
<body>
<h2 id="txt">This is a Sample Text</h2>
<br><br>
<button onclick="bigFont()">Big Font</button>
<button onclick="colorRed()">Red Text</button>
<button onclick="changeImg()">Change Image</button>
<button onclick="resetAll()">Reset</button>

<script>
function bigFont() {
document.getElementById("txt").style.fontSize = "32px";
}
function colorRed() {
document.getElementById("txt").style.color = "red";
}
function changeImg() {
document.getElementById("pic").src =
"https://via.placeholder.com/150/FF0000?text=New";
}
function resetAll() {
var t = document.getElementById("txt");
t.style.fontSize = "";
t.style.color = "";
document.getElementById("pic").src =
"https://via.placeholder.com/150";
}
</script>
</body>
</html>
```

Practical No. 44

Q. Write a JavaScript program to calculate simple interest, compound interest, and percentage using buttons and functions.

```
<!DOCTYPE html>
<html>
<head>
<title>Interest Calculator</title>
<style>
input { margin-bottom: 8px; padding: 5px; width: 200px; }
button { padding: 8px 15px; margin: 4px; cursor: pointer; }
#res { font-weight: bold; color: darkblue; margin-top: 10px; }
</style>
</head>
<body>
<h2>Interest Calculator</h2>
Principal (P): <input type="number" id="p"><br>
Rate (R %): <input type="number" id="r"><br>
Time (T yrs): <input type="number" id="t"><br>
<button onclick="calcSI()">Simple Interest</button>
<button onclick="calcCI()">Compound Interest</button>
<button onclick="calcPct()">Percentage</button>
<div id="res"></div>

<script>
function getVals() {
return {
p: parseFloat(document.getElementById("p").value),
r: parseFloat(document.getElementById("r").value),
t: parseFloat(document.getElementById("t").value)
};
}
function calcSI() {
var v = getVals();
var si = (v.p * v.r * v.t) / 100;
document.getElementById("res").innerHTML =
"Simple Interest = Rs. " + si.toFixed(2);
}
function calcCI() {
var v = getVals();
var ci = v.p * Math.pow((1 + v.r / 100), v.t) - v.p;
document.getElementById("res").innerHTML =
"Compound Interest = Rs. " + ci.toFixed(2);
}
function calcPct() {
var v = getVals();
document.getElementById("res").innerHTML =
v.r + "% of " + v.p + " = " + ((v.p * v.r) / 100).toFixed(2);
}
</script>
</body>
</html>
```

Practical No. 45

Q. Design a PHP script to display student information and browser details using predefined server variables.

```
<!DOCTYPE html>
<html>
<head><title>Student Info</title></head>
<body>
<h2>Student Information</h2>
<?php
$name = "Shreyas Patil";
$roll = "22CE101";
$branch = "Computer Engineering";
$sem = "Semester IV";

echo "Name: $name <br>";
echo "Roll No: $roll <br>";
echo "Branch: $branch <br>";
echo "Semester: $sem <br><br>";

echo "<h3>Browser and Server Details</h3>";
echo "IP Address: " . $_SERVER['REMOTE_ADDR'] . "<br>";
echo "Browser: " . $_SERVER['HTTP_USER_AGENT'] . "<br>";
echo "Server: " . $_SERVER['SERVER_SOFTWARE'] . "<br>";
echo "Current Date: " . date("d-m-Y") . "<br>";
?>
</body>
</html>
```

Practical No. 46

Q. Write a PHP program to accept user feedback using POST method and display the entered details in formatted output.

```
<!DOCTYPE html>
<html>
<head><title>Feedback Form</title></head>
<body>
<h2>User Feedback</h2>
<form method="POST">
Name: <input type="text" name="name" required><br><br>
Email: <input type="email" name="email" required><br><br>
Rating:
<select name="rating">
<option>Excellent</option>
<option>Good</option>
<option>Average</option>
<option>Poor</option>
</select><br><br>
Comments:<br>
<textarea name="comments" rows="4" cols="40"></textarea><br><br>
<input type="submit" name="btn" value="Submit Feedback">
</form>
<hr>
<?php
if (isset($_POST['btn'])) {
$n = htmlspecialchars($_POST['name']);
$e = htmlspecialchars($_POST['email']);
$r = htmlspecialchars($_POST['rating']);
$c = htmlspecialchars($_POST['comments']);
echo "<h3>Feedback Received!</h3>";
echo "<b>Name:</b> $n <br>";
echo "<b>Email:</b> $e <br>";
echo "<b>Rating:</b> $r <br>";
echo "<b>Comments:</b> $c <br>";
}
?>
</body>
</html>
```

Practical No. 47

Q. Implement a PHP program for array operations such as sorting, searching, and finding maximum value.

```
<!DOCTYPE html>
<html>
<head><title>Array Operations</title></head>
<body>
<h2>Array Operations</h2>
<?php
$marks = array(75, 90, 55, 88, 62, 45, 95);
echo "Original Array: " . implode(" ", $marks) . "<br>";

sort($marks);
echo "Sorted (Asc): " . implode(" ", $marks) . "<br>";

rsort($marks);
echo "Sorted (Desc): " . implode(" ", $marks) . "<br>";

echo "Max Value: " . max($marks) . "<br>";
echo "Min Value: " . min($marks) . "<br>";

$search = 88;
$found = in_array($search, $marks);
echo "Search $search: " . ($found ? "Found" : "Not Found") . "<br>";
echo "Total Elements: " . count($marks) . "<br>";
?>
</body>
</html>
```

Practical No. 48

Q. Create a PHP script to store product names and prices in an associative array and display them in tabular format.

```
<!DOCTYPE html>
<html>
<head><title>Product List</title></head>
<body>
<h2>Product Price List</h2>
<?php
$products = array(
    "Laptop" => 55000,
    "Smartphone" => 18000,
    "Tablet" => 25000,
    "Headphones" => 3500,
    "Keyboard" => 1200
);
echo "<table border='1' cellpadding='8'>";
echo "<tr><th>Product</th><th>Price (Rs.)</th></tr>";
foreach ($products as $name => $price) {
echo "<tr><td>$name</td><td>$price</td></tr>";
}
echo "</table>";
?>
</body>
</html>
```

Practical No. 49

Q. Create an HTML page to display a tourist place description using headings, paragraphs, images, and unordered lists.

```
<!DOCTYPE html>
<html>
<head><title>Tourist Place</title></head>
<body>
<h1>Mahabaleshwar - Queen of Hill Stations</h1>


<h2>About</h2>
<p>Mahabaleshwar is a hill station in Satara district, Maharashtra.
It is famous for strawberries, scenic viewpoints, and pleasant weather.</p>

<h2>Top Attractions</h2>
<ul>
<li>Venna Lake</li>
<li>Elephant Head Point</li>
<li>Pratapgad Fort</li>
<li>Mapro Garden</li>
<li>Arthur's Seat</li>
</ul>

<h2>How to Reach</h2>
<p>Nearest city: <b>Pune</b> (120 km). Best reached by road.</p>
</body>
</html>
```

Practical No. 50

Q. Design a web page containing a photo gallery, hyperlinks to educational websites, and a weekly class schedule using tables.

```
<!DOCTYPE html>
<html>
<head><title>Class Schedule</title></head>
<body>
<h2>Photo Gallery</h2>




<h2>Educational Websites</h2>
<a href="https://nptel.ac.in" target="_blank">NPTEL</a> |
<a href="https://coursera.org" target="_blank">Coursera</a> |
<a href="https://geeksforgeeks.org" target="_blank">GeeksforGeeks</a>

<h2>Weekly Schedule</h2>
<table border="1" cellpadding="8">
<tr><th>Day</th><th>9-10</th><th>10-11</th><th>11-12</th><th>2-4 (Lab)</th></tr>
<tr><td>Monday</td> <td>DBMS</td><td>DM</td> <td>IOT</td> <td>DBMS Lab</td></tr>
<tr><td>Tuesday</td> <td>COM</td> <td>DBMS</td><td>DM</td> <td>WD Lab</td></tr>
<tr><td>Wednesday</td><td>IOT</td> <td>COM</td> <td>DBMS</td><td>-</td></tr>
<tr><td>Thursday</td> <td>DM</td> <td>IOT</td> <td>COM</td> <td>COM Lab</td></tr>
<tr><td>Friday</td> <td>DBMS</td><td>DM</td> <td>IOT</td> <td>-</td></tr>
</table>
</body>
</html>
```

Practical No. 51

Q. Create an online job application form with fields for applicant name, email, mobile number, qualification, experience, and submit button.

```
<!DOCTYPE html>
<html>
<head><title>Job Application</title></head>
<body>
<h2>Online Job Application</h2>
<form>
Applicant Name: <input type="text" name="name" required><br><br>
Email: <input type="email" name="email" required><br><br>
Mobile Number: <input type="tel" name="mobile" required><br><br>
Qualification:
<select name="qual">
<option>B.E.</option>
<option>M.E.</option>
<option>MBA</option>
<option>Diploma</option>
</select><br><br>
Experience (years): <input type="number" name="exp" min="0"><br><br>
<input type="submit" value="Apply Now">
<input type="reset" value="Clear">
</form>
</body>
</html>
```

Practical No. 52

Q. Apply CSS styles using inline, internal, and external methods to design a registration form and styled alert messages.

```
<!DOCTYPE html>
<html>
<head>
<title>Styled Registration</title>
<style>
/* Internal CSS */
body { font-family: Arial; background: #eef; }
.form-container {
background: white; width: 350px; padding: 25px;
margin: 30px auto; border-radius: 8px;
box-shadow: 0 0 10px rgba(0,0,0,0.1);
}
input[type="text"], input[type="email"] {
width: 100%; padding: 8px; margin-bottom: 10px;
border: 1px solid #ccc; border-radius: 4px;
}
.btn { background: #5cb85c; color: white; padding: 10px 20px;
border: none; border-radius: 4px; cursor: pointer; }
.btn:hover { background: #449d44; }
.alert-box {
background: #d4edda; border: 1px solid #c3e6cb;
padding: 10px; border-radius: 4px; margin-top: 10px;
}
</style>
</head>
<body>
<div class="form-container">
<!-- Inline CSS heading -->
<h2 style="color:#5cb85c; text-align:center;">Register</h2>
<input type="text" placeholder="Full Name">
<input type="email" placeholder="Email Address">
<button class="btn">Register</button>
<div class="alert-box">
Registration form ready. Fill details and submit.
</div>
</div>
</body>
</html>
```


Practical No. 53

Q. Develop a responsive restaurant website homepage using Bootstrap Navbar, Carousel, Cards, and Grid Layout.

■ *FIX: Bootstrap JS bundle was MISSING — carousel prev/next buttons did NOT work. Added JS bundle + navbar toggler.*

```
<!DOCTYPE html>
<html>
<head>
<title>Restaurant Website</title>
<meta name="viewport" content="width=device-width, initial-scale=1">
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
<div class="container">
<span class="navbar-brand fw-bold">Spice Garden</span>
<button class="navbar-toggler" type="button"
data-bs-toggle="collapse" data-bs-target="#navMenu">
<span class="navbar-toggler-icon"></span>
</button>
<div class="collapse navbar-collapse" id="navMenu">
<ul class="navbar-nav ms-auto">
<li class="nav-item"><a class="nav-link" href="#">Home</a></li>
<li class="nav-item"><a class="nav-link" href="#">Menu</a></li>
<li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
</ul>
</div>
</div>
</nav>

<!-- Carousel -->
<div id="menuCarousel" class="carousel slide" data-bs-ride="carousel">
<div class="carousel-inner">
<div class="carousel-item active">
<div class="bg-warning text-dark text-center p-5">
<h2>Delicious Food Awaits!</h2>
</div>
</div>
<div class="carousel-item">
<div class="bg-success text-white text-center p-5">
<h2>Fresh Ingredients Daily</h2>
</div>
</div>
</div>
<button class="carousel-control-prev" data-bs-target="#menuCarousel" data-bs-slide="prev">
<span class="carousel-control-prev-icon"></span>
</button>
<button class="carousel-control-next" data-bs-target="#menuCarousel" data-bs-slide="next">
<span class="carousel-control-next-icon"></span>
</button>
</div>

<div class="container mt-4">
<h3 class="text-center">Our Menu</h3>
<div class="row">
<div class="col-md-4 mb-3">
```

```
<div class="card shadow">
<div class="card-body text-center">
<h5>Starters</h5>
<p>Soup, Salad, Spring Rolls</p>
</div>
</div>
</div>
<div class="col-md-4 mb-3">
<div class="card shadow">
<div class="card-body text-center">
<h5>Main Course</h5>
<p>Biryani, Dal Tadka, Paneer</p>
</div>
</div>
</div>
<div class="col-md-4 mb-3">
<div class="card shadow">
<div class="card-body text-center">
<h5>Desserts</h5>
<p>Gulab Jamun, Ice Cream</p>
</div>
</div>
</div>
</div>
</div>

<!-- Bootstrap JS is REQUIRED for Carousel and Navbar toggler -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Practical No. 54

Q. Write a JavaScript program to validate a feedback form including email validation, comment length, and mandatory field checking.

```
<!DOCTYPE html>
<html>
<head>
<title>Feedback Validation</title>
<script>
function validate(e) {
var name = document.getElementById("name").value;
var email = document.getElementById("email").value;
var comments = document.getElementById("comments").value;
var msg = "";

if (name == "")
msg += "Name is required.\n";
if (!/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email))
msg += "Invalid email format.\n";
if (comments.length < 10)
msg += "Comments must be at least 10 characters.\n";

if (msg != "") { alert(msg); e.preventDefault(); }
else alert("Feedback submitted successfully!");
}
</script>
</head>
<body>
<h2>Feedback Form</h2>
<form onsubmit="validate(event)">
Name: <input type="text" id="name"><br><br>
Email: <input type="text" id="email"><br><br>
Comments:<br>
<textarea id="comments" rows="4" cols="35"></textarea><br><br>
<input type="submit" value="Submit Feedback">
</form>
</body>
</html>
```

Practical No. 55

Q. Create a web page that changes text style, image size, and background color dynamically using JavaScript events.

```
<!DOCTYPE html>
<html>
<head><title>Dynamic Events</title></head>
<body>
<h2 id="heading">Sample Heading</h2>
<br><br>
<button onclick="boldText()">Bold Text</button>
<button onclick="bigImage()">Big Image</button>
<button onclick="changeBg()">Yellow Background</button>
<button onclick="resetAll()">Reset</button>

<script>
function boldText() {
document.getElementById("heading").style.fontWeight = "bold";
document.getElementById("heading").style.color = "darkblue";
}
function bigImage() {
document.getElementById("img").width = 250;
}
function changeBg() {
document.body.style.backgroundColor = "yellow";
}
function resetAll() {
var h = document.getElementById("heading");
h.style.fontWeight = "normal";
h.style.color = "black";
document.getElementById("img").width = 150;
document.body.style.backgroundColor = "white";
}
</script>
</body>
</html>
```

Practical No. 56

Q. Write a JavaScript program to perform area calculations of circle, rectangle, and triangle using functions and buttons.

```
<!DOCTYPE html>
<html>
<head>
<title>Area Calculator</title>
<style>
input { margin-bottom: 8px; padding: 5px; width: 180px; }
button { padding: 8px 14px; margin: 4px; cursor: pointer; }
#res { font-weight: bold; color: darkgreen; margin-top: 10px; }
</style>
</head>
<body>
<h2>Area Calculator</h2>
Value 1: <input type="number" id="v1" placeholder="Radius / Length / Base"><br>
Value 2: <input type="number" id="v2" placeholder="Width / Height (if needed)"><br>
<button onclick="circleArea()">Circle Area</button>
<button onclick="rectArea()">Rectangle Area</button>
<button onclick="triArea()">Triangle Area</button>
<div id="res"></div>

<script>
function circleArea() {
var r = parseFloat(document.getElementById("v1").value);
document.getElementById("res").innerHTML =
"Circle Area = " + (Math.PI * r * r).toFixed(2);
}
function rectArea() {
var l = parseFloat(document.getElementById("v1").value);
var w = parseFloat(document.getElementById("v2").value);
document.getElementById("res").innerHTML =
"Rectangle Area = " + (l * w).toFixed(2);
}
function triArea() {
var b = parseFloat(document.getElementById("v1").value);
var h = parseFloat(document.getElementById("v2").value);
document.getElementById("res").innerHTML =
"Triangle Area = " + (0.5 * b * h).toFixed(2);
}
</script>
</body>
</html>
```

Practical No. 57

Q. Design a PHP script to display customer details and current date/time information from the server.

```
<!DOCTYPE html>
<html>
<head><title>Customer Details</title></head>
<body>
<h2>Customer Details</h2>
<?php
$customers = array(
array("name" => "Shreyas", "email" => "shreyas@mail.com", "city" => "Pune"),
array("name" => "Radha", "email" => "radha@mail.com", "city" => "Mumbai"),
array("name" => "Krishna", "email" => "krishna@mail.com", "city" => "Nashik")
);
echo "<table border='1' cellpadding='8'>";
echo "<tr><th>Name</th><th>Email</th><th>City</th></tr>";
foreach ($customers as $c) {
echo "<tr><td>{$c['name']}</td><td>{$c['email']}</td><td>{$c['city']}</td></tr>";
}
echo "</table><br>";
echo "<b>Server Date:</b> " . date("d-m-Y") . "<br>";
echo "<b>Server Time:</b> " . date("h:i:s A") . "<br>";
?>
</body>
</html>
```

Practical No. 58

Q. Write a PHP program to accept student registration details using POST method and display registration success message.

```
<!DOCTYPE html>
<html>
<head><title>Student Registration</title></head>
<body>
<h2>Student Registration Form</h2>
<form method="POST">
Name: <input type="text" name="name" required><br><br>
Email: <input type="email" name="email" required><br><br>
Password: <input type="password" name="pass" required><br><br>
Branch:
<select name="branch">
<option>Computer Engineering</option>
<option>Mechanical</option>
<option>Civil</option>
</select><br><br>
<input type="submit" name="btn" value="Register">
</form>
<hr>
<?php
if (isset($_POST['btn'])) {
$name = htmlspecialchars($_POST['name']);
$branch = htmlspecialchars($_POST['branch']);
echo "<h3 style='color:green;'>Registration Successful!</h3>";
echo "Welcome <b>$name</b>! You are registered in <b>$branch</b>.";
}
?>
</body>
</html>
```

Practical No. 59

Q. Implement a PHP program for numeric operations such as finding factorial, palindrome number, and reverse number.

```
<!DOCTYPE html>
<html>
<head><title>Numeric Operations</title></head>
<body>
<h2>Numeric Operations</h2>
<?php
$num = 5;

// Factorial
$fact = 1;
for ($i = 1; $i <= $num; $i++) $fact *= $i;
echo "Factorial of $num = $fact <br>";

// Reverse
$rev = (int) strrev((string) $num);
echo "Reverse of $num = $rev <br>";

// Palindrome check
if ($num == $rev) echo "$num is a Palindrome <br>";
else echo "$num is NOT a Palindrome <br>";

// Check 121
$num2 = 121;
$rev2 = (int) strrev((string) $num2);
echo "<br>Reverse of $num2 = $rev2 <br>";
if ($num2 == $rev2) echo "$num2 is a Palindrome <br>";
else echo "$num2 is NOT a Palindrome <br>";
?>
</body>
</html>
```

Practical No. 60

Q. Create a PHP script to store book names and authors in an associative array and display them in table format.

```
<!DOCTYPE html>
<html>
<head><title>Book List</title></head>
<body>
<h2>Book List</h2>
<?php
$books = array(
    "Let Us C" => "Yashavant Kanetkar",
    "Head First Java" => "Kathy Sierra",
    "DBMS Concepts" => "Silberschatz",
    "The Lean Startup"=> "Eric Ries",
    "Clean Code" => "Robert C. Martin"
);
echo "<table border='1' cellpadding='8'>";
echo "<tr><th>Book Name</th><th>Author</th></tr>";
foreach ($books as $book => $author) {
echo "<tr><td>$book</td><td>$author</td></tr>";
}
echo "</table>";
?>
</body>
</html>
```